Apm 1110 FA 8

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

① 10 RED 8 BLACK 7 GREEN

4 RANDOMLY CHOSEN $N = 25$

$X \rightarrow$ NO. OF BALLS THAT ARE OF THE CHOSEN COLOR

a) $P(X = 0)$

PROBABILITY OF CHOOSING EACH COLOR $= \frac{1}{3}$

PROBABILITY THAT RED IS NOT CHOSEN IN 4 SELECTIONS:

$25 - 10 = 15$

$$P(X=0 \mid R) = \frac{\binom{15}{4}}{\binom{25}{4}} = \frac{1365}{12650}$$

$$\binom{15}{4} = \frac{15!}{4!(15-4)!} = 1365$$

$$\binom{25}{4} = \frac{25}{4!(25-4)!} = 12650$$

BLACK: $25 - 8 = 17$

$$P(X=0 \mid B) = \frac{\binom{17}{4}}{\binom{25}{4}} = \frac{2380}{12650}$$

$$\binom{17}{4} = \frac{17!}{4!(17-4)!} = 2380$$

GREEN: $25 - 7 = 18$

$$P(X=0 \mid G) = \frac{\binom{18}{4}}{\binom{25}{4}} = \frac{3060}{12650}$$

$$\binom{18}{4} = \frac{18!}{4!(18-4)!} = 3060$$

$$P(X=0) = \sum P(X=0 \mid \text{colors})\, P(\text{colors})$$

$$P(X=0) = \frac{1}{3}\left(\frac{1365 + 2380 + 3060}{12650}\right)$$

$$= \frac{1}{3} \cdot \frac{1361}{2530}$$

$$= \frac{1361}{7590}$$

$$\boxed{P(X=0) = 0.1793}$$

b) $P(X_i = 1)$, $i = 1, 2, 3, 4$

$$X_i = \begin{cases} 1, & \text{if the } i^{th} \text{ ball selected is of chosen color} \\ 0, & \text{otherwise} \end{cases}$$

$$P(X_i = 1 \mid \text{RED}) = \frac{10}{25}$$

$$P(X_i = 1 \mid \text{BLACK}) = \frac{8}{25}$$

$$P(X_i = 1 \mid \text{GREEN}) = \frac{7}{25}$$

$$P(\text{colors}) = \frac{1}{3}$$

$$P(X_i = 1) = \sum P(X_i = 1 \mid \text{colors})\, P(\text{colors})$$

$$= \frac{1}{3}\left(\frac{10 + 8 + 7}{25}\right)$$

$$= \frac{1}{3} = 0.3333\ldots$$

$P(X_1 = 1) = 0.3333$ $P(X_3 = 1) = 0.3333$

$P(X_2 = 1) = 0.3333$ $P(X_4 = 1) = 0.3333$

c) $E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4]$

$E[X_i] = P(X_i = 1)$ 1.6)

$P(X_i = 1) = \frac{1}{3}$

$$E[X] = \frac{1}{3} + \frac{1}{3} + \frac{1}{3} + \frac{1}{3} = \frac{4}{3}$$

$$\boxed{E[X] = 1.3333}$$

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② YULE-SIMON DISTRIBUTION

$$P(X=n) = \frac{4}{n(n+1)(n+2)}, \quad n \geq 1$$

a) SHOW THAT $\sum_{n=1}^{\infty} P(X=n) = 1$

PRECEDING IS PMF

$P(X=n) > 0$

$$= \frac{4}{n(n+1)(n+2)} \quad n \geq 1$$

If $n=1$ $$= \frac{4}{1(1+1)(1+2)}$$

$0 < \frac{4}{6}$ NON NEGATIVE

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right] \times 4$$

$$\frac{4}{n(n+1)(n+2)} = 2\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$\sum_{n=1}^{\infty} P(X=n) = 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

$$= 2\left[\frac{1}{1(1+1)} - \frac{1}{(1+1)(1+2)}\right]$$

$$= 2\left[\left(\frac{1}{2} - \frac{1}{2(3)}\right) + \left(\frac{1}{2(2+1)} - \frac{1}{(2+1)(2+2)}\right) + \left(\frac{1}{3(3+1)} - \frac{1}{(3+1)(3+2)}\right) + \cdots\right]$$

$$= 2\left(\frac{1}{2}\right) = 1$$

$$\boxed{\sum_{n=1}^{\infty} P(X=n) = 1}$$

PMF ✓

b) $E[X] = 2$

$$E[X] = \sum_{n=1}^{\infty} n P(X=n)$$

$$= \sum_{n=1}^{\infty} n\left(\frac{4}{n(n+1)(n+2)}\right)$$

$$= \sum_{n=1}^{\infty} \frac{4n}{n(n+1)(n+2)}$$

$$= 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}$$

$$\frac{1}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2} \quad \times (n+1)(n+2)$$

$$1 = A(n+2) + B(n+1)$$

$n = -1$

$1 = A(-1+2) + B(-1+1)$

$1 = A(1)$

$n = -2$

$1 = A(-2+2) + B(-2+1)$

$1 = B(-1)$

$-1 = B$ $\qquad \frac{1}{n+1} - \frac{1}{n+2}$

$$E[X] = 4\sum_{n=1}^{\infty}\left(\frac{1}{n+1} - \frac{1}{n+2}\right)$$

$$= 4\left[\left(\frac{1}{1+1} - \frac{1}{1+2}\right) + \left(\frac{1}{2+1} - \frac{1}{2+2}\right) + \left(\frac{1}{3+1} - \frac{1}{3+2}\right) + \cdots\right]$$

$$= 4\left(\frac{1}{2}\right) = 2$$

$$\boxed{E[X] = 2}$$

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c) $E[X^2] = \infty$

$$= \sum_{n=1}^{\infty} n^2 \left(\frac{4}{n(n+1)(n+2)}\right)$$

$$= \sum_{n=1}^{\infty} \frac{4n^2}{n(n+1)(n+2)}$$

$$= 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}$$

$$\frac{n}{(n+1)(n+2)} = \frac{A}{n+1} + \frac{B}{n+2} \qquad -\frac{1}{n+1} + \frac{2}{n+2}$$

$n = A(n+2) + B(n+1)$

$n+1$

$-1 = A(-1+2)$

$-1 = A(1)$

$n = -2$

$-2 = B(-2+1)$

$-2 = -B$

$2 = B$

$$E[X^2] = 4\sum_{n=1}^{\infty} \left(-\frac{1}{n+1} + \frac{2}{n+2}\right)$$

$$= 4\left[\left(-\frac{1}{1+1} + \frac{2}{1+2}\right) + \left(-\frac{1}{2+1} + \frac{2}{2+2}\right) + \left(-\frac{1}{3+1} + \frac{2}{3+2}\right) + \left(-\frac{1}{4+1} + \frac{2}{4+2}\right) + \dots\right]$$

$$= 4\left[\left(-\frac{1}{2} + \frac{2}{3}\right) + \left(-\frac{1}{3} + \frac{2}{4}\right) + \left(-\frac{1}{4} + \frac{2}{5}\right) + \left(-\frac{1}{5} + \frac{2}{6}\right) + \dots\right]$$

$$= 4\left[\left(\frac{2}{3} + \frac{2}{4} + \frac{2}{5} + \frac{2}{6} + \dots\right) - \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots\right)\right]$$

BOTH CAN BE COMPARED TO HARMONIC SERIES

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots = \infty$$

$\therefore E[X^2] = \infty$

(3) PDF $f(x) = \begin{cases} a + bx^2, & 0 \le x \le 1 \\ 0, & \text{otherwise} \end{cases}$

IF $E[x] = \frac{3}{5}$. FIND VALUES OF a, b.

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_0^1 a + bx^2\,dx = 1$$

$$\int_0^1 a\,dx + \int_0^1 bx^2\,dx = 1$$

$$a[x]_0^1 + b\left[\frac{x^3}{3}\right]_0^1 = 1$$

$1 - 0 = 1$ $\qquad \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}$

$$a + \frac{b}{3} = 1 \qquad (a)$$

$$E[X] = \int_{-\infty}^{\infty} x f(x)\,dx$$

$$\frac{3}{5} = \int_0^1 x(a + bx^2)\,dx$$

$$= \int_0^1 ax + bx^3\,dx$$

$$= \int_0^1 ax\,dx + \int_0^1 bx^3\,dx$$

$$= a\left[\frac{x^2}{2}\right]_0^1 + b\left[\frac{x^4}{4}\right]_0^1$$

$\frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2}$ $\qquad \frac{1^4}{4} - \frac{0^4}{4} = \frac{1}{4}$

$$\frac{3}{5} = \frac{a}{2} + \frac{b}{4} \qquad (b)$$

(a) $a + \frac{b}{3} = 1$

$a = 1 - \frac{b}{3}$

(b) $\frac{a}{2} + \frac{b}{4} = \frac{3}{5}$

$\frac{1 - \frac{b}{3}}{2} + \frac{b}{4} = \frac{3}{5}$

$\frac{1}{2} - \frac{b}{6} + \frac{b}{4} = \frac{3}{5}$

$-\frac{2b}{12} + \frac{3b}{12} = \frac{3}{5} - \frac{1}{2}$

$\frac{b}{12} = \frac{6}{10} - \frac{5}{10}$

$\left(\frac{b}{12} = \frac{1}{10}\right) 12$

$\boxed{b = \frac{12}{10} = \frac{6}{5}}$

$a = 1 - \frac{6/5}{3}$

$= 1 - \frac{6}{15}$

$\boxed{a = \frac{3}{5}}$

(4) X → RANDOM VARIABLE
μ → EXPECTED VALUE
σ^2 → VARIANCE

$Y = \frac{X-\mu}{\sigma}$ FIND EXPECTED VALUE $E[Y]$ AND VARIANCE $Var(Y)$

EXPECTED VALUE

$E[aX+b] = aE[X]+b$

$E[Y] = E\left[\frac{X-\mu}{\sigma}\right]$

$= \frac{1}{\sigma} E[X-\mu]$

$= \frac{1}{\sigma}(E[X]-\mu)$ $\quad E[X]=\mu$

$= \frac{1}{\sigma}(\mu-\mu)$

$\boxed{E[Y] = 0}$

VARIANCE

$Var(aX+b) = a^2 Var(X)$

$Y = \frac{X-\mu}{\sigma}$

$= \frac{X}{\sigma} - \frac{\mu}{\sigma}$ $\quad a = \frac{1}{\sigma}$ $\quad b = -\frac{\mu}{\sigma}$

$Var(Y) = \left(\frac{1}{\sigma}\right)^2 Var(X)$ $\quad Var(X) = \sigma^2$

$= \frac{1}{\sigma^2}(\sigma^2)$

$\boxed{Var(Y) = 1}$

(5) X, Y → INDEPENDENT POISSON RANDOM VARIABLES W PARAMETERS
$\lambda_1 = 2 \quad \lambda_2 = 3$

LET $W = X + Y$

a) DERIVE MOMENT GENERATING FUNCTION (MGF) $M_W(t)$ OF W

$M_X(t) = e^{\lambda(e^t-1)}$

X $\lambda = 2$ $\quad$ Y $\lambda = 3$

$M_X(t) = e^{2(e^t-1)}$ $\quad M_Y(t) = e^{3(e^t-1)}$

$W = X + Y$

$M_W(t) = M_X(t) M_Y(t)$

$= e^{2(e^t-1)} \cdot e^{3(e^t-1)}$

$= e^{2(e^t-1)+3(e^t-1)}$

$\boxed{M_W(t) = e^{5(e^t-1)}}$

b) PROBABILITY DISTRIBUTION & PARAMETERS OF W

(a) $M_W(t) = e^{5(e^t-1)}$ $\quad \lambda = 5$

$M_X(t) = e^{\lambda(e^t-1)}$

$\boxed{W \sim P(5)}$

c) $E[W]$ EXPECTED VALUE $\quad t = 0$

(a) $M_W(t) = e^{5(e^t-1)}$

$M'_W(t) = e^{5(e^t-1)}(5e^t)$

$= 5e^t e^{5(e^t-1)}$

$E[W] = M'_W(0)$ $\quad t = 0$

$= 5e^0 e^{5(e^0-1)}$

$\boxed{E[W] = 5}$

$E[W^2]$ SECOND MOMENT

$M'_W(t) = 5e^t e^{5(e^t-1)}$ $\quad u'v + uv'$

$M''_W(t) = 5e^t e^{5(e^t-1)} + 5e^t e^{5(e^t-1)} 5e^t$

$= 5e^t e^{5(e^t-1)} + 25e^{2t} e^{5(e^t-1)}$

$E[W^2] = M''_W(0)$ $\quad t = 0$

$= 5e^0 e^{5(e^0-1)} + 25e^{2(0)} e^{5(e^0-1)}$

$= 5 + 25$

$\boxed{E[W^2] = 30}$

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